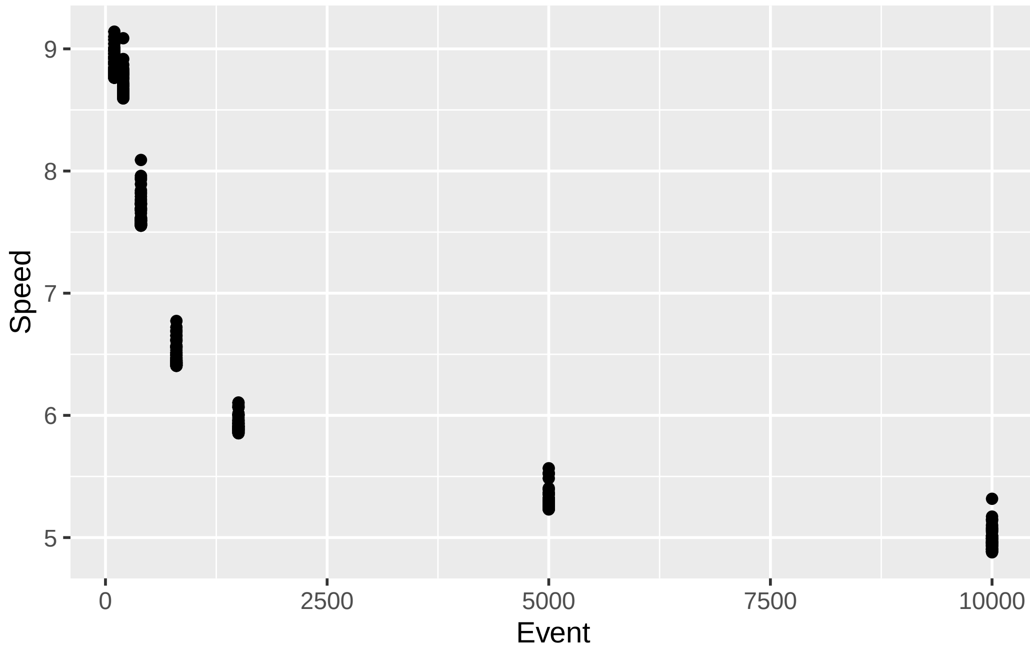
2025 Women's Track Running: Fixing Non-Linear Data Transformation | SOLUTIONS

```
library(tidyverse)
library(ggplot2)

track <- read.csv("WomensTrack2025.csv")
```

1. Make a scatterplot using distance (Event) to predict speed

```
ggplot(track, aes(x = Event, y = Speed)) +
  geom_point()
```



1.1. What do you notice? (think of model conditions)

The data does not match a linear trend. There is clear curvature in the data. Speed seems to at first, dramatically decrease as distance increases, but the speed at which it decreases slows as distance continues to grow.

1.2. Given the context from the first paragraph and/or your background knowledge of the sport, give a possible explanation for what you noticed.

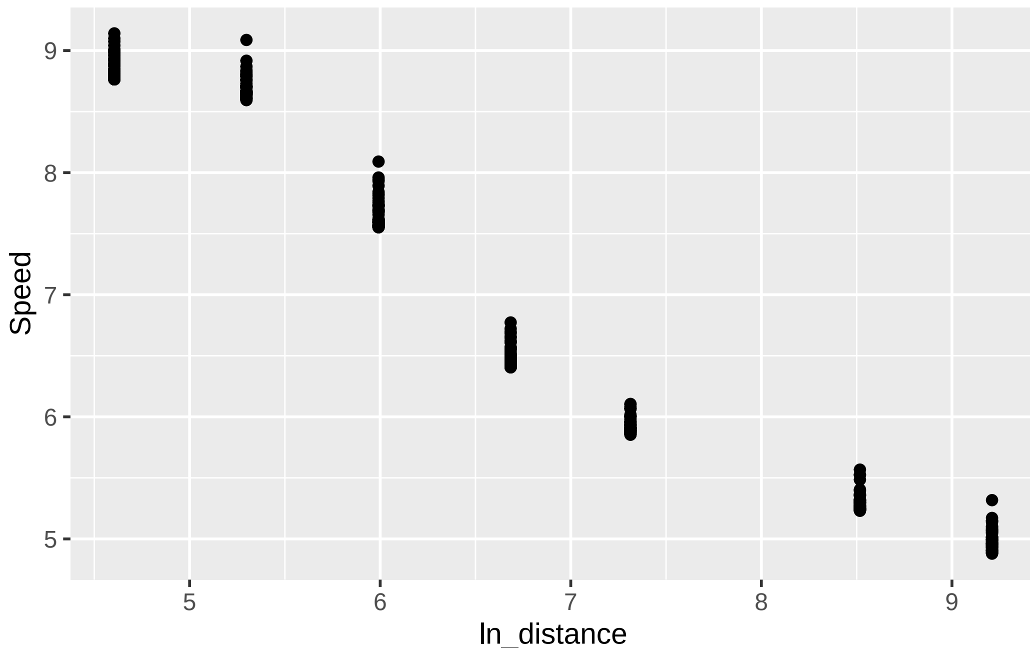
Answers may vary. Students may point to the fact that distance runners have to run for longer, and therefore cannot most likely maintain their fastest possible speed for this time period. The sprinters, on the other hand, must do exactly that, as they only have a short period of time to get and keep the lead.

2. Based on the model's issue, apply a transformation to the data to make a linear model more appropriate.

```r
track <- track |> mutate(ln_distance = log(Event))
```

3. Make a scatterplot showing the transformed relationship.

```
ggplot(track, aes(x = ln_distance, y = Speed)) +
  geom_point()
```



4. Define a model using the transformed data and print a summary.

```
mod_track <- lm(Speed ~ ln_distance, track)
summary(mod_track)
```

```
Call:
lm(formula = Speed ~ ln_distance, data = track)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.55817 -0.33019 -0.00245  0.31186  0.84663 

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.12022    0.08295  158.18   <2e-16 ***
ln_distance -0.92106    0.01190  -77.42   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3448 on 349 degrees of freedom
Multiple R-squared:  0.945, Adjusted R-squared:  0.9448
F-statistic:  5995 on 1 and 349 DF,  p-value: < 2.2e-16
```

5. Interpret in context, the coefficient on the transformed distance term.

As the natural log of distance increases by 1, speed will, on average, decrease by 0.92 meters/second.